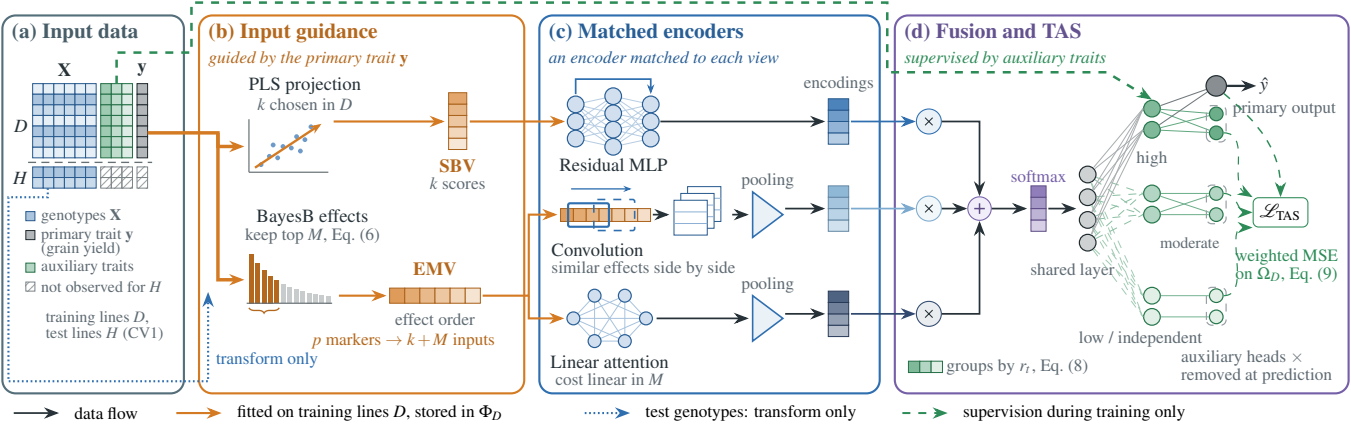

(a) Input data
X
y
D
H
genotypes X
primary trait y (grain yield)
auxiliary traits
not observed for H
training lines D, test lines H (CV1)
(b) Input guidance
guided by the primary trait y
PLS projection
k chosen in D
SBV
k scores
BayesB effects
keep top M, Eq. (6)
EMV
effect order
p markers → k + M inputs
transform only
(c) Matched encoders
an encoder matched to each view
encodings
Residual MLP
pooling
Convolution
similar effects side by side
pooling
Linear attention
cost linear in M
(d) Fusion and TAS
supervised by auxiliary traits
softmax
shared layer
ŷ
primary output
high
moderate
low / independent
ℒTAS
weighted MSE on ΩD, Eq. (9)
auxiliary heads × removed at prediction
groups by rt, Eq. (8)
data flow
fitted on training lines D, stored in ΦD
test genotypes: transform only
supervision during training only